

Beverly Xu

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EDUCATION

Harvey Mudd College | B.S. in Engineering

Relevant coursework

Digital Electronics · Engineering Design and Development · Linear Algebra · Multivariable Calculus · Physics
Mechanics and E&M · Special Relativity

PUBLICATIONS

"cHyRRT and cHySST: Two Motion Planning Tools for Hybrid Dynamical Systems," IEEE CASE, August 2025

ROBOTICS EXPERIENCE

Electrical and Hardware Member, Harvey Mudd RoboSub Team, Claremont, CA

September 2025 - Present

Designing an AUV capable of underwater missions and acoustic localization for the annual RoboSub competition.

- Designed three-stage 25-40 kHz hydrophone signal amplifier with 20,000-120,000x adjustable gain, DC bias stages to preserve signal through single-rail op-amps and protect Teensy input, and low-pass filter to reduce noise from adjacent motors. Designed in KiCAD and simulated using LTSpice.
- Converted signal processing PCB from 2 to 4 layers, added testing areas, created bill of materials using KiCAD.
- Designed torpedo launcher using SolidWorks FEA and flow simulation to model force distribution and fluid dynamics.
- prototyped using 3D printer, manufactured 2-inch diameter launcher using Fusion CAM and DPM SX3P Bed Mill.

Software Lead, Mechanical Science and Engineering Robotics Team, Saratoga, CA

August 2021 - June 2025

Implementing robotic control systems for the FIRST Robotics Competition; 2-year Worlds division finalists/semifinalists.

- Implemented control systems (feedforward, current-limiting, PID) in Java using encoders, IR gates, limit switches, etc.
- Troubleshoot robot breakdowns in CAN loops, actuators, power distribution panel, sensor firmware, application-level software.

RESEARCH EXPERIENCE

Intern, Lab for Investigations of Local Air Quality, Claremont, CA

September 2025 - Present

- Prototyped CO2 level indicator with I2C interface to SCD41 sensor; designed enclosure in SolidWorks using edit-in-context assembly workflow.
- Wrote Linux, BASH, and IGOR scripts for Raspberry Pi and Windows devices to transfer data realtime to Firebase from PurpleAir and meteorological sensors running on legacy middleware.

Intern, UC Santa Cruz, Hybrid Systems Lab, Santa Cruz, CA

August 2023 - August 2025

- Implemented two motion planners for hybrid dynamical systems (collision-resilient quadcopter, bouncing ball, etc.) in the C++ Open Motion Planning Library; wrote and presented paper at IEEE CASE 2025.
- Designed C++ tests to validate motion plans, Python scripts for visualization, and VM image.

PERSONAL PROJECTS

Analog Alarm Clock, Claremont, CA

April 2026 - May 2026

Single motor-actuated second, minute, and hour hands, with WiFi and Bluetooth connectivity for custom alarm sounds.

- Designed custom gear box in SolidWorks with 1:60:360 gear ratios, evaluated dynamics using motion study.
- Breadboarded circuit with Raspberry Pi Pico 2W, 5V DC buck converter, lightweight stepper motor, and switches.

Regenerative Dance Floor Tile, Saratoga, CA

January 2025 - May 2025

Spring-loaded tile which converts mechanical energy from steps to chemical energy stored in a battery.

- Collaborated with a team of 3 to develop a step mechanism, converting bidirectional linear motion to rotational motion, and capable of bearing a max load of 150 lbs, to backdrive motor and generate electricity.
- Converted generated AC to DC using diode bridge rectifier; troubleshoot issues with conversion and battery-charging circuit.

TECHNICAL SKILLS

Mechanical/Manufacturing: SolidWorks, Fusion, Siemens NX, 3D printing, CAM

Electrical: circuitry simulation, KiCAD, LTSpice, Raspberry Pi, Arduino, soldering

Software: C++, MATLAB, Python, LaTeX, Java, BASH, Linux, motion planning algorithms, control systems

HONORS

Society of Women Engineers Motorola Foundation Scholarship 2025

Alltricity Foundation Scholarship 2025

Harvey Mudd Merit Scholarship 2024